



Little Engineer

A calm, offline train game for a four-year-old

Storybook railway, four big buttons, nothing can go wrong.

This is the settled design and the phased build plan. It records decisions, not options — where something is genuinely undecided it appears under *Open questions* at the end.

PHASE 0 DELIVERABLE

Settled design & build plan

TARGET

Google Pixel Tablet, offline PWA

STATUS

Ready to begin Phase 1

The game

What it is, and who it is for

Make train go → watch train → blow whistle → stop somewhere interesting → trigger something fun → keep going.

Playable with no reading, no explanation, no account, no internet, no ads, no purchases, no menus to navigate and no railway knowledge.

Player profile

This is the most important input to the design, and it is specific rather than generic. Watching him in *Train Simulator PRO*, he gravitates to three things:

- 1 Blowing the whistle**
Constantly, often more than he drives.
- 2 Stopping at stations**
The arrival, the ritual, the pause.
- 3 Changing cameras**
He enjoys watching as much as driving.

He does **not** chase speed. He is not a throttle child; he is a conductor of a play-mat railway. Three decisions follow directly:

- **No speed control.** GO / STOP only. A speed slider would add complexity serving an appetite he doesn't have. Revisit only through parent settings, Phase 8.
- **The whistle is a primary control, not a garnish.** Equal visual weight to GO and STOP, always on screen, always responsive, and the world answers it.
- **The camera button is a first-class control,** not a settings item. Flipping views is play.

Settled decisions

Name	Little Engineer
Core gameplay	Drive a train round a continuous loop; stop at stations; whistle at everything; watch it go
Controls	GO, STOP, WHISTLE, CAMERA. Four buttons. Nothing else on screen
Camera	One big button cycles three fixed views: Wide, Follow, Trackside
World	One continuous loop, several stations, visible as a play-mat layout. No end, no fail
View style	Storybook 3/4 — angled overhead, a toy railway on a table

Activities	Passengers board · station comes alive · whistle gets a response
Track interaction	Fully automatic. No junctions in the MVP (deferred to Phase 5)
Difficulty	Generous stopping zone with gentle auto-assist. He cannot miss
Failure states	None whatsoever. Overshooting simply means going round again
Rewards	None in the MVP. Optional sticker book deferred to Phase 7
Audio	Sounds and animation only. No speech, in any phase
Sessions	Endless free play. Start instantly, stop anytime, nothing is lost
Parent controls	Deferred to Phase 8, behind a long-press gesture
Engines	One engine in the MVP. Engine selection deferred to Phase 6
Artwork	Original engines and scenery, designed for this project
Delivery	Installable offline PWA on a free static host

The railway

One closed loop, six things to arrive at

A single continuous loop, laid out so that most of it is visible at once in Wide view — the feel of looking down at a wooden train set on the floor. Six events evenly spaced around it means something happens roughly every fifteen seconds: close enough that the gaps never feel empty, far enough apart that the journey between them still counts as driving.

Stations

Three in the MVP. Each has a distinct silhouette and colour so it is identifiable with zero reading.

STATION	IDENTITY	SIGNATURE MOMENT
The Sheds	Red brick, home, where play begins	Doors open as the engine leaves
The Harbour	Blue, boats, water, cranes	A boat passes and sounds its horn back
The Farm	Green, barn, animals, hay	Sheep look up when whistled at

Trackside features

So that something happens while simply driving.

- **A tunnel** — whistle inside and it echoes.
- **A level crossing** — gates lower, a car waits, the driver waves.
- **A bridge** — water underneath, a boat may be passing.

The track is a closed circuit. The engine always follows it. There is no steering, no switching, and no way to leave the rails.

Why a loop. There is no finish, so there is nothing to lose and no “again?” to negotiate. He stops when he is done, and the tablet can be taken away mid-lap without interrupting anything.

Controls and screen layout

Designed landscape-first. A four-year-old holds a tablet like a steering wheel, so the primary controls sit in the bottom corners where thumbs already are — never along the top.

GO	Bottom-left, green, 164 px. Largest target on screen.
STOP	Bottom-right, red, 164 px. Opposite corner, so the two can never be confused.
WHISTLE	Bottom-centre, amber, 124 px. Works stopped or moving, no cooldown, never greys out.
CAMERA	Top-left, 88 px. Cycles Wide → Follow → Trackside.
Minimum target	96 px anywhere in the game — double the adult minimum.

Every press answers three ways at once: **the button moves, the world reacts, and a sound plays** — within one frame, every time. Immediate feedback matters more than realism.

Camera, arrivals and sound

The three things he actually does

Camera

One button, three fixed views, cycling in a fixed order so the sequence becomes predictable.

Wide	The whole layout, train small, best for watching it go round. The default.
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Follow	Close behind and slightly above the engine, track moving beneath. The “driving” view.
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Trackside	A low fixed shot near the rails; the train enters, passes, and leaves the frame.
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No free camera, no pinch-zoom, no drag. He cannot get the view stuck underground or aimed at the sky. Transitions are smooth glides, never cuts.

The station moment

Stations are his favourite thing, so this is the most polished interaction in the game.

Approach	Within range of a platform the engine eases off automatically. The station brightens slightly — a soft cue that something is possible here, with no arrow and no text.
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Stopping	The good-stop zone is deliberately very wide. Tapping STOP anywhere near the platform counts as a perfect arrival. He feels responsible for the stop; he effectively cannot fail it.
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Arrival	Three things happen together: passengers walk out and board; the station comes alive (signal, flag, bell, lamps); and whistling gets an answer — people wave, birds scatter, the stationmaster blows back.
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Leaving	Tapping GO sets off again. Nothing is required of him; he may sit at a station whistling for five minutes, and that is a perfectly valid way to play.
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Overshooting	Not a failure. The train simply carries on and comes round again.
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Audio

No speech, in any phase. Sounds and animation carry everything — there are no lines to record or re-record, nothing becomes irritating on the two hundredth hearing, and it works regardless of language or reading ability.

- **Whistle** — warm, clear, satisfying, instant. The single most-heard sound in the game; worth getting genuinely right before anything else.
- **Chuffing** — speed-matched, soft, seamless loop; fades to a gentle idle hiss when stopped.
- **World** — birds, water, a distant church bell, wind, sheep, the crossing bell.
- **Arrival** — a short, unforced chime. Celebratory, not a fanfare, because he will hear it hundreds of times.

Explicitly not in this game

Recorded here so it doesn't creep back in later.

accounts · logins · internet requirement · ads · purchases · currency · XP · levels · daily rewards · streaks · timers · countdowns · scores · lives · derailment · collisions · red-signal penalties · tutorials · text instructions · menus more than one tap deep · analytics · cloud saves · servers · any ongoing cost

The build

Eleven phases, each one tested with him before the next begins

Feature count is not the measure. **Independent, unprompted play is.** After every phase, stop and watch him use it.

PHASE 1

Toddler UX prototype ← start here

- One engine, one small loop, one station, simple scenery.
- GO / STOP / WHISTLE / CAMERA, at full size.
- Basic PWA install, so he can start it himself from the home screen.
- No progression, no saved data, no extra content.

Deliverable He picks up the tablet and plays without being told how. Nothing else counts.

PHASE 2

Train-driving foundation

- Track-following along a spline, with smooth acceleration and braking curves.
- The station-proximity assist and the generous stop zone.
- The camera rig and its three fixed views.
- Written so additional engines and routes reuse it unchanged.

Deliverable A stable, reusable train controller and track system.

PHASE 3

The complete loop

- Three stations, tunnel, level crossing, bridge.
- The full station moment: boarding, the station waking up, the whistle response.

Deliverable The first version that is genuinely the game rather than a test.

PHASE 4

Activity system

- Small toys added one at a time, never chores: freight wagons to collect and drop off, a findable animal, an engine wash, a branch to tap off the track, a visiting engine.
- Every one of them ignorable. Missing an activity must never punish or block him.

Deliverable A small library of reusable, toddler-friendly interactions.

PHASE 5

Expand the railway

- New areas that connect to the existing world rather than replacing it.
- Junction choice with two very large arrow buttons — but only if Phase 4 shows he wants agency.
- Do not build a large map merely because the architecture allows it.

Deliverable Several connected, polished stretches of railway.

PHASE 6

More engines

- A second and third engine reusing the same controller.
- Differing only in appearance, whistle, top speed and small animation flourishes.
- Visual selection screen, no text.

Deliverable A picture-only engine chooser plus reusable engine definitions.

PHASE 7

Gentle progression

- An optional local sticker book recording places visited and things seen.
- Nothing can be lost, nothing expires, nothing pressures him to return.

Deliverable An optional, local-only discovery book.

PHASE 8

Parent settings

- A long-press gesture opens a hidden panel, well away from ordinary play.
- Assist strength, speed control on/off, junctions on/off, which activities are enabled.
- At four, the default stays GO / STOP / WHISTLE.

Deliverable A hidden parent panel stored locally — how the game grows with him from four to six.

PHASE 9

Polish

- Only for things he actually uses repeatedly.
- Weather, day and night, more scenery, more sounds, more character reactions.

Deliverable A polished version of the proven game, not a feature-heavy simulator.

PHASE 10

Pixel Tablet packaging

- Full offline service worker, bundled assets, local storage, fullscreen, home-screen icon.
- No server, no accounts, no analytics, no ongoing cost.

Deliverable The finished installable offline PWA.

First milestone

Deliberately tiny, and the only thing Phase 1 has to achieve

- 1 Tap the icon on the Pixel Tablet home screen.
- 2 It opens fullscreen, instantly, offline.
- 3 One engine sits on one loop.
- 4 Tap **GO** — it moves.
- 5 Tap **STOP** — it stops.
- 6 Tap **WHISTLE** — a proper whistle, immediately.
- 7 Tap **CAMERA** — the view changes.
- 8 Arrive at one station — something visibly happens.
- 9 Keep driving as long as he likes.

If that holds his attention and he can use it unaided, continue. **If not, change the interaction before adding any content.**

Steps 1 and 2 pull basic PWA install forward from Phase 10 into Phase 1. This is deliberate: testing the game in a browser tab with an address bar is not testing the real thing, and “can he start it himself?” is part of what Phase 1 measures.

Technical approach

Renderer **Three.js**, stylised low-poly, angled camera for the 3/4 storybook view. Chosen because Wide, Follow and Trackside are genuinely different viewpoints — faking those in 2D means three separate sets of artwork; in 3D the camera gives them for free. Low-poly geometry is also buildable in code, which keeps the art achievable.

Stack	Vanilla TypeScript and Vite. No game engine, no framework, no state library — the game is small, and a framework would be more moving parts than the game itself.
Storage	localStorage only. No database, no server, no network calls of any kind at runtime.
Performance	Locked 60 fps. Tensor G2 is comfortable with low-poly, but the screen is 2560 × 1600 — so render at a capped device pixel ratio rather than native.

Structure

Content is separated from engine logic from the start — the one piece of architecture worth having early, because it is what makes Phases 5 and 6 cheap.

```
src/  
  engine/    track following, camera rig, input, audio  
  content/   engines, stations, scenery, sounds  
  ui/        the four buttons
```

The engine layer never references a specific character, station or theme.

Target, assets and loose ends

The tablet it runs on, the art it uses, and what is still undecided

Pixel Tablet targets

Screen	10.95", 2560 × 1600, 16:10
CSS viewport	~1280 × 800, landscape
Orientation	Landscape locked. Portrait is not supported or designed for
Render resolution	Capped device pixel ratio (~1.5×) — framerate over sharpness
Minimum touch target	96 px; primary controls 164 px
Install	PWA, fullscreen standalone, no browser chrome
Offline	Total. Flight mode must make no difference

Artwork and separation

This is a private family project, but the code should not depend on that staying true.

All engine appearances, names, faces, scenery and sounds are original work created for this project, and live entirely in `src/content/`. The engine layer never references a specific character, station or theme. The practical consequence: the game can be shared, shown to other parents, or published without anything needing to be unpicked.

Open questions

- 1 **The engine's name.** Worth asking him — a four-year-old naming his own engine is free ownership of the game. The nameplate on the tank is blank until then, and it is the only text in the entire game.
- 2 **Whistle sound source.** The most-used sound in the game deserves a deliberate choice: a real steam whistle recording, or something warmer and more toy-like.
- 3 **Where the build is hosted.** A free static host is decided; which one is not. It only matters for how updates reach the tablet.

The rule that governs all of it. Do not expand scope beyond the active phase. Prefer simple, replaceable implementations. Preserve working behaviour between phases. Favour reliability and immediate feedback over simulation realism. And after each phase, stop and test with the child before going on.